MILITARY SPECIFICATION

INDICATOR, PRESSURE, TYPE TB-1

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope This specification covers one type of hermetically sealed, remote indicating, voltage ratio type, oil pressure indicator designated Type TB-1.

2. APPLICABLE DOCUMENTS

2.1 Government Furnished Documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

Military

MIL-I-58098	Indicators, Pressure, Helicopter, Voltage Ratio Type, General Specification for
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic
MIL-T-58100	Transmitter, Pressure, Oil Voltage Ratio Type

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Requirements for Reliability Programs (for Systems and Equipments)
MIL-STD-831	Test Reports, Preparation of

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MS33585 Pointer, Dial, Standard Design of Aircraft Instrument
MS33639 Cases, Instrument, Clamp Mounted, Aircraft.

(Copies of specifications, standards, drawings, and publications required by suppliers ir connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction This specification makes provisions for preproduction testing. The requirements of MIL-I-58098 apply to this specification except as specified herein When these specifications conflict, this specification shall govern.

3.2 Design and construction Indicator shall be designed for use with a transmitter conforming to MIL-T-58100.

3.2.1 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.2.1.1 Indicator reliability. The indicator shall have a minimum acceptable MTBF (mean-time-between-failures) of 2,000 hours, at a confidence level of 90 percent, and a discrimination ratio of 1.5.

3.2.1.2 Indicator longevity. The indicator shall have an operating life span of 3,000 hours.

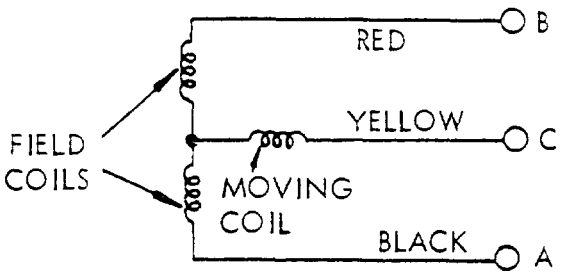
3.2.2 Case. The case dimensions shall conform to MS33639 for 2 inch size. Case length shall be 2 5/8 ± 3/8 inches.

3.2.3 Pointer. The pointer shall be in accordance with MS33585-8 except that the length shall be so that the tip will extend into the shortest graduation a distance approximately one third the length of that graduation.

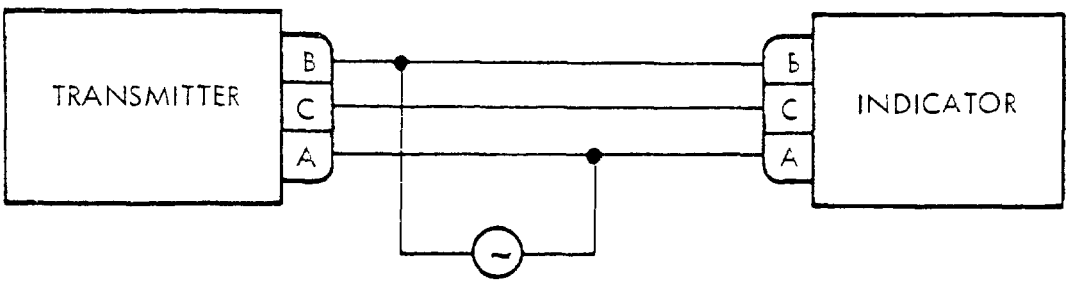
3.2.3.1 Pointer finish. Shaded portion of the pointer shall be finished in white, color No 37875 of FED-STD-595.

3.2.4 Dial. The dial shall conform to figure 1 The graduation mark for 45 psi shall be located on the horizontal centerline of the left side of the dial. A minimum diameter of 1 3/4 inches across the outside of graduation shall be maintained.

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INTERNAL WIRING

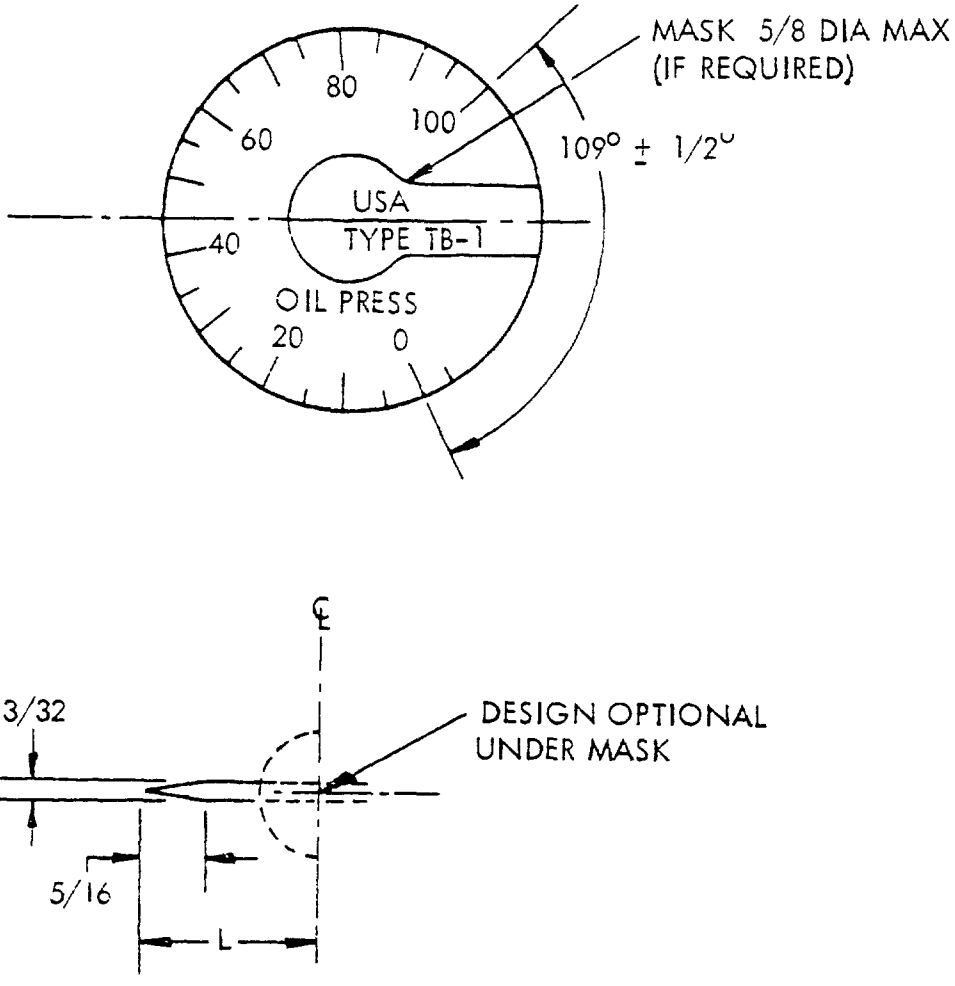


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POWER SUPPLY

EXTERNAL WIRING

Figure 2. Wiring Diagram

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ± 1/64

Figure 1. Dial and Pointer

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3.2.4.1 Marking configuration. The following marking shall be finished in white, color No. 37875 of FED-STD-595. Dimensions shall be as follows:

	Height or Length Inch + 1/64	Width of Line or Graduation Inch +0.000
Numerals 0, 20, 40, 60 80 and 100	5/32	0.025
10-pound graduations	3/16	0.031
5-pound graduations	3/32	0.020
Lettering-OIL PRESS	1/8	0.025
Lettering-PSI	1/8	0.025
Mask reference lines	---	0.020
Pointer reference mark	1/16	0.015

3.2.4.2 Identification markings. The markings US ARMY and TYPE TB-1 shall be permanently marked in letters 1/16 inch in height. Unless otherwise specified, black color No. 37038 of FED-STD-595 shall be used.

3.2.5 Electrical wiring. Internal and external wiring shall conform to figure 2 and shall be connected to receptable pins as shown thereon. Internal wiring shall be color coded

3.2.6 Weight. Weight of the indicator shall not exceed 0.50 pounds.

3.2.7 Bearings When synthetic ruby or sapphire bearings are used, they shall conform to MIL-B-27497.

3.3 Performance.

3.3.1 Tolerances. Indicators shall operate within the tolerances specified in Tables I and II.

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*TABLE I. Temperature Scale Error

Test Pressure PSI	Room Temperature ± PSI	High and Low Temp Exposure High and Low Temperature ± PSI
0	0.50	1.5
10	0.50	1.5
20	0.50	1.5
30	0.75	1.75
40	0.75	1.75
50	0.75	1.75
60	0.75	1.75
70	0.75	1.75
80	0.75	1.75
90	0.75	1.75
100	0.75	1.75

*TABLE II. Position Friction and Vibration Error

Test	Tolerance ± PSI
Position error at 50 psi	1.0
Friction	
Room temperature	1.0
High temperature	1.0
Low temperature	1.5
Vibration	
Pointer oscillation	1.0
Pointer variation	1.0

4. Quality Assurance Provisions.

4.1 Responsibility for Inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

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4.2 General requirements. The inspection and testing of indicators shall be in accordance with MIL-I-58098 and as specified herein. The indicator shall operate at all times within the tolerances specified in para 3.3.1.

4.3 Preproduction testing.

4.3.1 Preproduction tests. The preproduction tests shall consist of all tests specified herein and in MIL-I-58098.

4.3.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831.

4.3.3 Test samples. The preproduction test samples shall consist of ten indicators representative of the production equipment. Three of the samples shall be subjected to all the tests except reliability and the other seven shall be subjected to the reliability tests only. Samples shall be identified with the manufacturer's part number and all additional information required by the procuring activity.

4.4 Test methods

4.4.1 Test attitude. Unless otherwise specified, the indicator shall be tested with the dial rotated as shown in figure 1 and the plane of the dial at a 20° angle from the vertical. (Top of the dial tilting toward the rear of the instrument).

4.4.2 Reliability. Unless otherwise specified, seven indicators as specified in para 4.3.3 shall be subjected to the following reliability test.

4.4.2.1 Procedure. The test equipment and fixtures shall include provisions for vibration and temperature cycling at periodic intervals specified herein. Throughtout this test, input signals from 0 to 85 percent of full scale and return shall be applied to the indicator at a rate of 2 to 3 cycles (minimum and maximum) per minute Test shall begin when rated power is applied. After completion of 4 hours operation with rated power on (at least 480 cycles of operation), the power shall be turned off and remain off for a period of 10 minutes. The indicator shall be vibrated in accordance with MIL-STD-781, test level C, for at least 10 minutes of every hour of ON time. The temperature cycles shall be conducted in the following sequence: 4 hours at -20°C, 14 minutes for temperature change to +50°C at a minimum rate of 5° per minute; 4 hours at +50°C,

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14 minutes for temperature change to -20°C; and 4 hours at -20°C. All malfunctions shall be recorded and the test interrupted until repairs are made. At each 250 hour cycling interval, the indicator shall be checked for scale error. Scale error reading shall not exceed 150 percent of room temperature tolerance in table I. No recalibration of units may be performed during this test The test may be interrupted for weekend shutdown The contractor shall prepare a report in accordance with MIL-STD-785 which will contain test procedures, details of malfunctions, and corrective actions taken.

4.4.3 Longevity Three indicators which have completed the reliability test specified in 4.4.2.1 shall continue the testing through a period of 3000 hours for each indicator Time accumulated during reliability test shall count toward this total time Accept-reject criteria shall not be applicable.

4.4.4 Vibration. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, equipment category (C) procedure 1, part 1 (Table 514.1-III) and curve M (Figure 514.1.3).

4.4.4.1 A standard transmitter circuit shall be adjusted to provide voltages corresponding to an indication of approximately midscale on the indicator. Pointer oscillation and variation shall be monitored throughout the test and shall not exceed the tolerances specified in Table II

4.4.4.2 The indicator shall then be subjected to a vibration search having peak-to-peak amplitude of .3 inches through a frequency range of 5 to 25 cps, and then to a vibration which produces 10g through a frequency range of 25 to 50 cps During these vibration searches, the pointer shall not vibrate more than ±2.5 psi, and the hairspring shall not cause an electrical short circuit The searches will be made on the 3 mutally perpendicular axes A constant 50 psi signal shall be applied during this test. Upon completion of this vibration test, the indicator shall be subjected to the scale error test at room temperature and scale errors shall not exceed the tolerances specified in Table I.

5. Preparation for delivery

5.1 Preparation for delivery shall be in accordance with MIL-I-58098 and the instructions of the procuring activity. (See 6.2)

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6. Notes

6.1 Intended use. The Type TB-1 indicators covered by this specification are intended for use in conjunction with pressure transmitters in accordance with specification MIL-T-58100 in indicating remotely the pressure in oil systems of aircraft/helicopter engine.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B will not be conducted

c. Levels of packaging and packing required (See 5.1)

6.2.1 Contract data requirements A report of all failures occuring during preproduction testing will be submitted to the procurring activity within 2 working days.

6.2.2 Test report. The contractor will submit three copies of the report specified in 4.3.2 to the procurring activity for approval.

Custodian:

ARMY-AV

Preparing Activity.

ARMY-AV
PR. NO. 6620-A005

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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